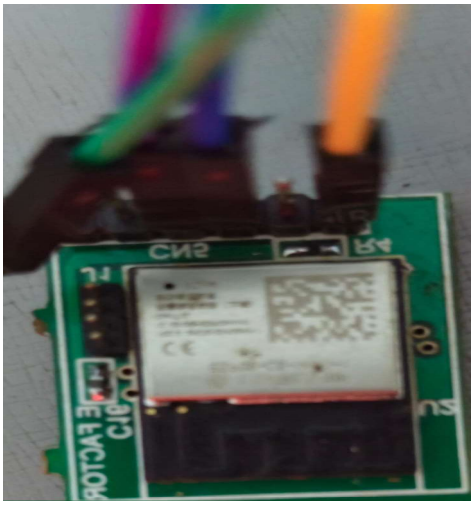
E-Factor SDK — Quick Start (ESP32-C6, Windows)

1) Hardware

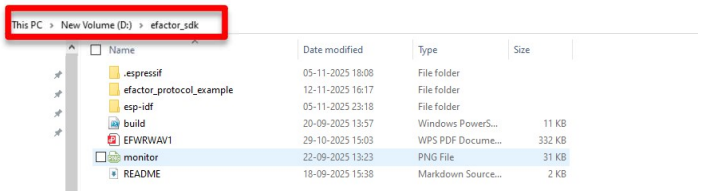
ESP32-C6 module with breakout board.

Ensure proper wiring as shown in the E-Factor IoT Module document (includes pins required for programming and Boot Interface sw1/sw2, and UART logging).



2) Software Setup

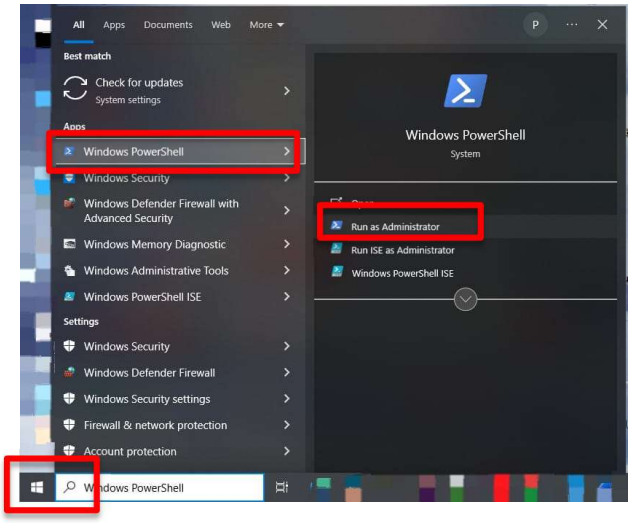
1. Download the 'efactor_sdk' zip file.
2. Extract it directly to Disk D:, so the folder becomes D:\efactor_sdk.



3) Open PowerShell in Administrator Mode

Press the Windows key, type 'PowerShell', right-click 'Windows PowerShell', and select 'Run as Administrator'.

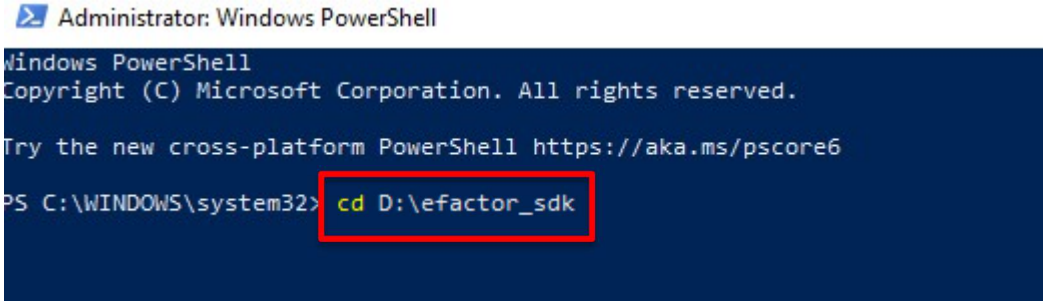
If PowerShell is not available, please install it for Windows.



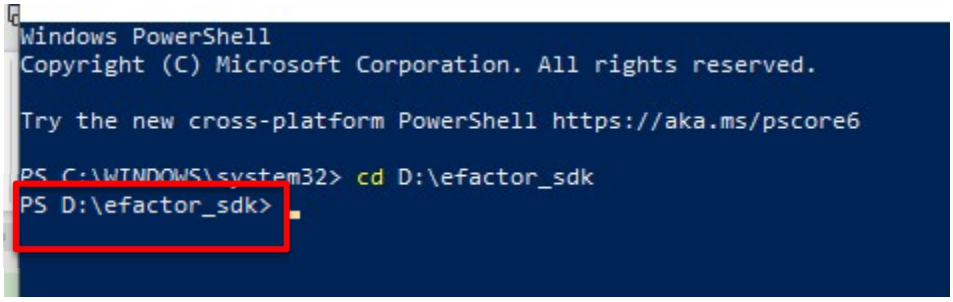
4) Navigate to the SDK Folder

Enter the following command:

cd D:\efactor_sdk



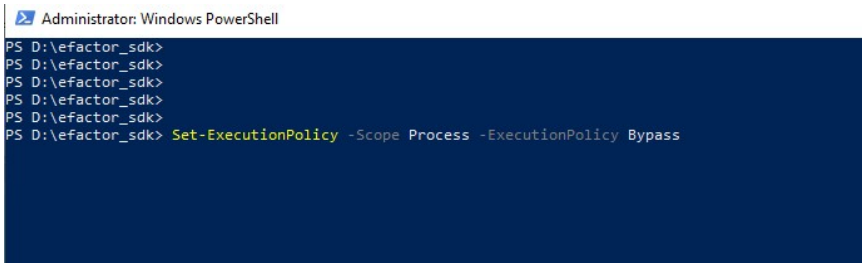
Press Enter

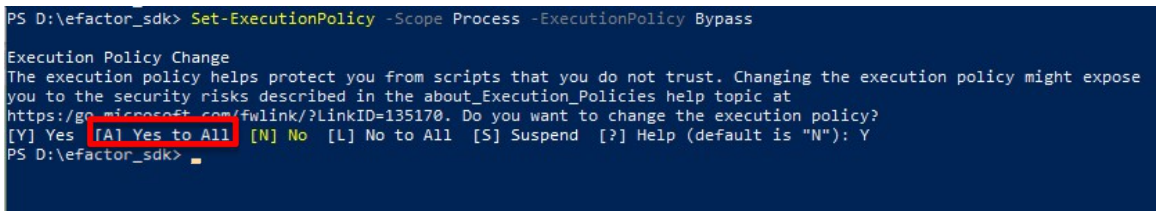


5) Allow Script Execution for the Session

Run this command:

Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass



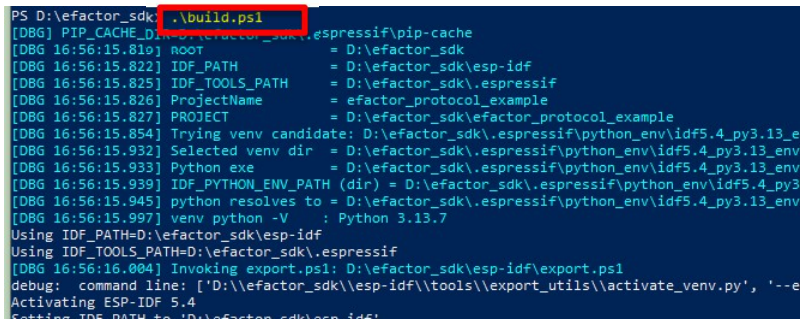


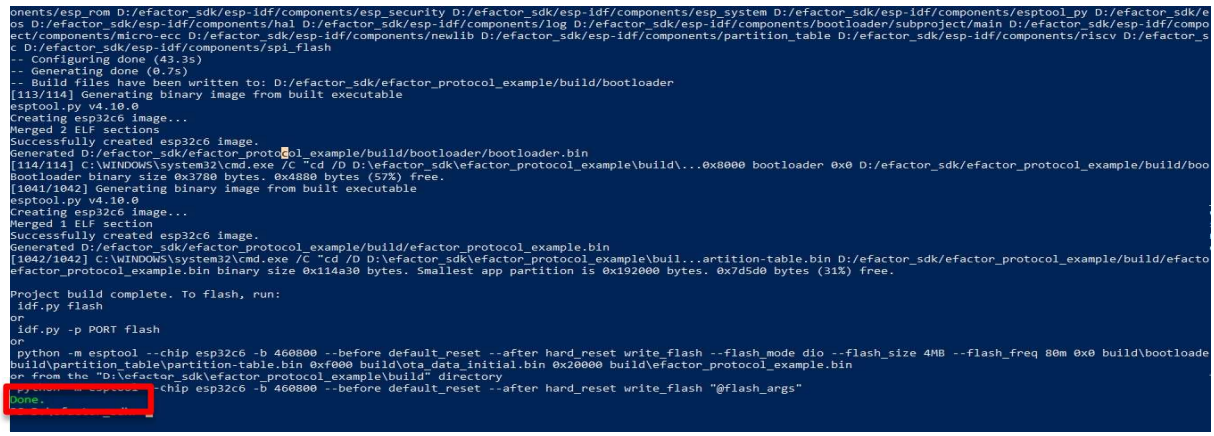
When prompted, type 'A' to confirm 'Yes to All'.

6) Build the Firmware

Execute:

.\build.ps1



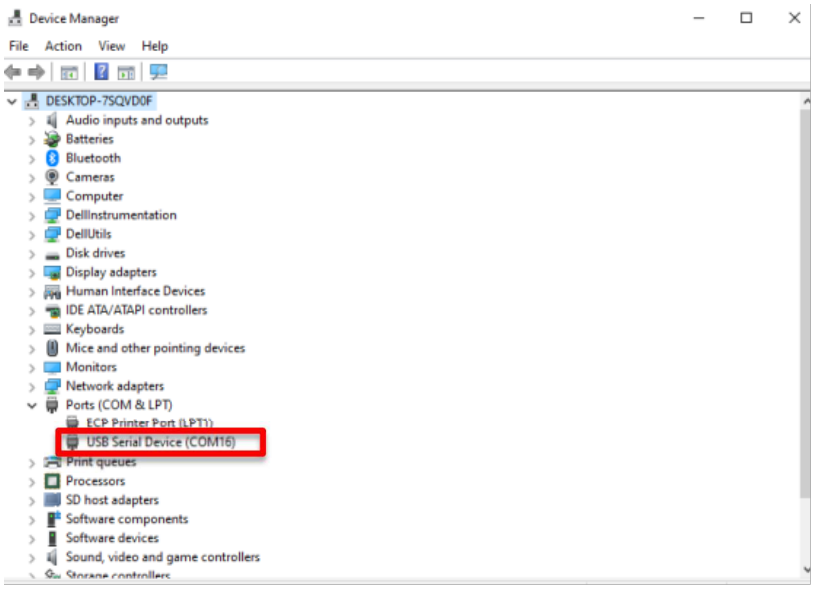


If the build is successful, a 'Done' message will appear.

7) Connect the Board and Identify COM Port

Plug in your ESP32-C6 board.

Open Device Manager → Ports (COM & LPT) → note your COM port number (e.g., COM16).



8) Flash and Monitor

Use the following command (replace COM16 with your actual port):

.\build.ps1 -Flash -Port COM16 -Monitor

The command will build, flash, and automatically start the UART monitor. When flashing completes, you'll see **Done**. After you reset the module, the monitor reconnects and displays the main.c welcome message, device UID, and E-Factor cloud connectivity status.

